



Operating System

Lab 7

Fasiha Tariq – 53289

Submitted to:

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TASKS

1. Read the contents of file1.txt and create file2.txt with the same content using redirection only in a single command.

```
student@student-virtual-machine:~$ cat > file1.txt
Hello!
My name is Fasiha Tariq
^C
student@student-virtual-machine:~$ cat < file1.txt > file2.txt
student@student-virtual-machine:~$ cat file2.txt
Hello!
My name is Fasiha Tariq
student@student-virtual-machine:~$
```

2. Run date and append the result to a file named dates.log every time you execute the command.

```
student@student-virtual-machine:~$ date >> dates.log
student@student-virtual-machine:~$ cat dates.log
ت 29 ستمبر 2025 PKT و 14:41:15
student@student-virtual-machine:~$
```

3. Run a command that will produce both valid output and an error. Redirect standard output to ok.txt and errors to err.txt.

```
student@student-virtual-machine:~$ mkdir bin
student@student-virtual-machine:~$ ls /bin /fasiha > ok.txt 2> err.txt
student@student-virtual-machine:~$ cat err.txt
ls: cannot access '/fasiha': No such file or directory
student@student-virtual-machine:~$ cat ok.txt
/bin:
[
aa-enabled
aa-exec
aa-features-abi
aconnect
acpidbg
acpi_listen
add-apt-repository
addpart
addr2line
airscan-discover
alsabat
alsaloop
alsamixer
alsatplg
alsaucm
amidi
amixer
apg
apgbfm
aplay
aplaymidi
apport-bug
apport-cli
apport-collect
apport-unpack
appres
appstreamcli
apropos
apt
apt-add-repository
apt-cache
apt-cdrom
apt-config
aptdcon
apt-extracttemplates
apt-ftparchive
apt-get
apt-key
apt-mark
apt-sortpkgs
```

4. List the contents of the /bin directory and also try to list a non-existent directory /no_such_dir and redirect both the normal output and the error messages to be saved together in a single file called all.log in one line.

```
student@student-virtual-machine:~$ ls /bin /mishal > all.log 2>&1
student@student-virtual-machine:~$ cat all.log
ls: cannot access '/mishal': No such file or directory
/bin:
[
aa-enabled
aa-exec
aa-features-abi
aconnect
acpidbg
acpi_listen
add-apt-repository
addpart
addr2line
airscan-discover
alsabat
alsaloop
alsamixer
alsatplg
alsaucm
amidi
amixer
apg
apgbfm
aplay
aplaymidi
apport-bug
apport-cli
apport-collect
apport-unpack
appres
appstreamcli
apropos
apt
apt-add-repository
apt-cache
apt-cdrom
apt-config
aptdcon
apt-extracttemplates
apt-ftparchive
apt-get
apt-key
apt-mark
apt-sortpkgs
apturl
```

5. Run a command that reads from input.txt and writes sorted, unique lines into unique.txt using pipe operator.

```
student@student-virtual-machine:~$ cat > input.txt
car
bike
train
ship
bus
bicycle
car
tricycle
motorbikes
bus
^C
student@student-virtual-machine:~$ sort input.txt | cat > unique.txt
student@student-virtual-machine:~$ cat unique.txt
bicycle
bike
bus
bus
car
car
motorbikes
ship
train
tricycle
```

6. What would be the result of the following commands.

- `cat filename > new` // filename is name of file that exist at current path
- `who > new`
- `ls | sort -r`
- `ls | sort -r >> new`

```
student@student-virtual-machine:~$ cat > filename
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student@student-virtual-machine:~$ cat filename > new
student@student-virtual-machine:~$ who > new
student@student-virtual-machine:~$ ls | sort -r
Videos
unique.txt
Templates
task5.c
task5
task4.c
task4
task3.c
task3
task2.c
task2
task1.c
task1
SortLabNumeric5
sortlab5
snap
SE5.txt
SE51.txt
SE
sana.c
sana
SA
riphah.c
riphah
Public
Pictures
new
name.c
name
Music
labsort5
LabSort
labSort
lab5
lab4
input.txt
infinite.c
infinite
fruit.txt
file.txt
filename
file2.txt
```

```
student@student-virtual-machine:~$ ls | sort -r >> new
student@student-virtual-machine:~$ cat new
student  tty2          2025-09-23 13:24 (tty2)
Videos
unique.txt
Templates
task5.c
task5
task4.c
task4
task3.c
task3
task2.c
task2
task1.c
task1
SortLabNumeric5
sortlab5
snap
SE5.txt
SE51.txt
SE
sana.c
sana
SA
riphah.c
riphah
Public
Pictures
new
name.c
name
Music
labsort5
LabSort
labSort
lab5
lab4
input.txt
infinite.c
infinite
fruit.txt
file.txt
filename
file2.txt
file2
file1.txt
```

- Sorts the contents of fruits.txt in alphabetical order. Filters the sorted output to only display fruits that contain the substring "ap".

```
student@student-virtual-machine:~$ cat > fruit.txt
banana
apple
grape
apricot
mango
pineapple
orange
kiwi
papaya
student@student-virtual-machine:~$ sort fruit.txt | grep ap
apple
apricot
grape
papaya
pineapple
```